



Physiology of the Defecation Reflex

Physiology GIT Block · MEDucation by Mattin Majid

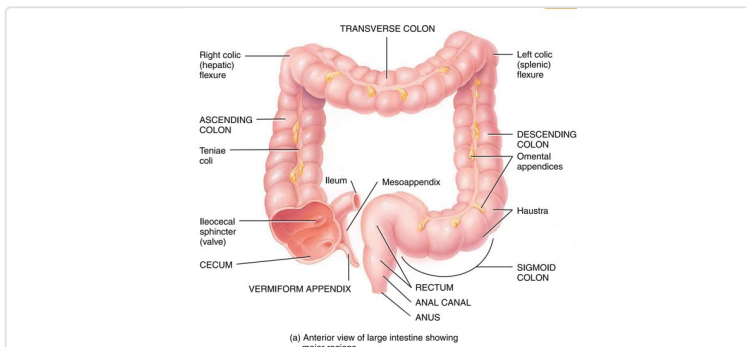
BEFORE YOU READ — ANSWER FROM MEMORY

1. What stimulates the defecation reflex, and where do the sensory signals go first?
2. Which anal sphincter is voluntary, and which is involuntary?
3. Why can infants not voluntarily control defecation?

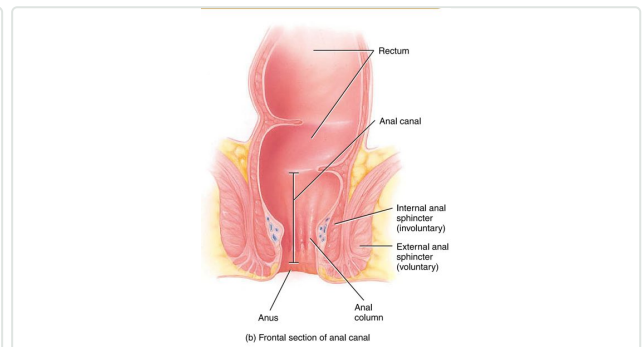
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 The Trigger

Once fecal material reaches the rectum, the resulting **distension of the rectal wall** stimulates **stretch receptors**, which initiates the defecation reflex — the elimination of feces from the rectum through the anus.



Large intestine — the fecal stream's path to the rectum and anal canal.



Internal (involuntary) vs external (voluntary) anal sphincter.

2 The Defecation Reflex — step by step

1. Sensory	Stretch receptors in the distended rectal wall send sensory impulses to the sacral spinal cord
2. Motor	Motor impulses travel back along parasympathetic nerves to the descending colon, sigmoid colon, rectum, and anus
3. Pressure	Contraction of the longitudinal rectal muscles shortens the rectum , raising intrarectal pressure
4. Internal sphincter opens	This pressure, plus voluntary diaphragm/abdominal contraction, plus parasympathetic stimulation, opens the internal anal sphincter (involuntary)
5. External sphincter decides	The external anal sphincter (voluntary) — if relaxed, defecation occurs; if constricted, defecation is postponed

3 If defecation is postponed

Feces back up into the sigmoid colon until the next wave of **mass peristalsis** re-stimulates the stretch receptors, recreating the urge to defecate.

4 Special notes

NEVER MISS: in **infants**, the defecation reflex causes automatic emptying of the rectum, because voluntary control of the external anal sphincter has not yet developed.

Normal bowel activity varies widely — from **2-3 movements per day** to **3-4 per week** — depending on diet, health, and stress.

5 Key mechanisms — the discriminator

Mechanism	Level	What it does
Intrinsic (enteric) reflex	Local — myenteric plexus	Rectal distension → local rectal contraction + internal sphincter relaxation
Parasympathetic reflex	Sacral cord, S2-S4	Amplifies rectal contraction and sphincter relaxation
Voluntary control	External anal sphincter + pelvic floor	Via the pudendal nerve
Central control	Cortex	Decides allow/withhold defecation → conscious relaxation of external sphincter + ↑ intra-abdominal pressure (Valsalva)

P Must-remember pearls

- Internal anal sphincter = **involuntary** (smooth muscle, parasympathetic); external anal sphincter = **voluntary** (skeletal muscle, pudendal nerve).
- The reflex has both an **intrinsic (myenteric)** component and a **parasympathetic (S2-S4)** component that amplifies it.
- Infants lack voluntary control of the external sphincter — defecation is purely reflexive until this develops.
- Postponing defecation doesn't cancel the reflex — feces back up into the sigmoid colon until the next mass peristalsis wave.
- Normal bowel habit range is wide: 3×/day to 3×/week are both normal.

Q Test yourself

1. A spinal cord injury above S2 would most likely result in:

- A. Complete loss of all defecation ability
- B. Loss of voluntary control, but reflex (involuntary) defecation may persist
- C. No change in bowel function
- D. Loss of internal sphincter tone only

2. Which nerve carries voluntary motor control to the external anal sphincter?

- A. Vagus nerve
- B. Pelvic splanchnic nerve
- C. Pudendal nerve
- D. Hypogastric nerve

3. Damage to spinal segments S2-S4 would most impair which component of the defecation reflex?

- A. Intrinsic (myenteric) reflex
- B. Parasympathetic reflex
- C. Voluntary cortical control
- D. Gastrocolic reflex

4. An infant's inability to voluntarily control defecation is explained by:

- A. Absence of stretch receptors in the rectum
- B. Undeveloped voluntary control of the external anal sphincter
- C. Absence of the internal anal sphincter
- D. Immature sacral spinal cord reflexes only

5. If a person voluntarily contracts the external anal sphincter during a defecation reflex, the fecal material:

- A. Is expelled regardless
- B. Backs up into the sigmoid colon until the next mass peristalsis wave
- C. Is reabsorbed by the rectum
- D. Triggers immediate internal sphincter closure permanently

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — B** — the intrinsic enteric reflex can still drive defecation without cortical/voluntary input, though continence is lost.
- 2 — C** — the pudendal nerve supplies voluntary (skeletal muscle) control of the external anal sphincter.
- 3 — B** — the parasympathetic component of the reflex arises from sacral segments S2-S4.
- 4 — B** — the reflex itself is present at birth; only voluntary external sphincter control is immature.
- 5 — B** — the reflex is deferred, not cancelled, until the next mass peristaltic wave restimulates the stretch receptors.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

